

Answer Key: Class Test 1 2026

1. A botanist is studying a new plant species. She observes two distinct pod shapes: plump and pinched. When she crosses two pure-breeding plants, one with plump pods and one with pinched pods, all the offspring (F1 generation) have plump pods. When these F1 plants are self-crossed, she gets an F2 generation with 73 plump pods and 24 pinched pods.

- a) Based on the F1 result, which phenotype is likely dominant? Explain in one sentence. (1 mark)

Plump pods are dominant because all F1 offspring from a cross between pure-breeding plump and pinched parents show the plump phenotype.

- b) The observed F2 numbers are approximately what classic phenotypic ratio? How many loci and how many alleles are likely to be involved in the inheritance of this trait? (1 mark)

The observed F2 numbers (73 plump : 24 pinched) are approximately a 3:1 phenotypic ratio. This suggests one locus with two alleles (dominant and recessive) controlling the trait.

- c) If the botanist randomly selects one F2 plant with plump pods, what is the probability (in fraction form) that it is pure-breeding (homozygous) for the plump trait? Show your reasoning. (1 mark)

In a monohybrid cross ($Aa \times Aa$), the F2 phenotypic ratio is 3 plump : 1 pinched.

Genotypic ratio: 1 AA : 2 Aa : 1 aa.

Plump phenotype includes: 1 AA (homozygous) + 2 Aa (heterozygous).

Probability that a randomly selected plump F2 plant is pure-breeding (homozygous) = $(1/3)$.

2. In some chickens, a single gene affects feather colour. A student performs the following cross:

- Parental Cross: A Grey-feathered chicken is crossed with a white-feathered chicken.
- Offspring (F1): Half the progeny are grey feather and half are white feathered.
- F1 Cross: Two grey-feathered chickens are crossed.
- F2 Offspring: The result is 25 black, 48 grey, and 27 white.

- a) Draw a genotype-phenotype map based on these results. Explain. (3 marks)

This can be explained by assuming the heterozygote having a different phenotype relative to the two homozygotes.

- Genotype → Phenotype map:

- BB → Black feathers
- Bb → Grey feathers
- bb → White feathers

Explanation:

- F1 cross (Grey × Grey = Bb × Bb) yields F2 ratio:

1 BB (black) : 2 Bb (grey) : 1 bb (white) → observed ≈ 25 : 48 : 27, which fits a 1:2:1 ratio.

3. In dogs, the allele for a wire-haired coat (W) is dominant to the allele for a smooth coat (w). A breeder has a wire-haired dog. To determine if this dog is homozygous (WW) or heterozygous (Ww), she decides to perform a cross.

- a) What specific genotype of dog *must* she mate her dog with for this test to be valid? (1 mark)

She must mate her wire-haired dog with a homozygous recessive (smooth-coated, ww) dog for a valid test cross.

- b) She can only afford to do the cross once, and it results in a litter of a single puppy, which has a wire-haired coat. The breeder concludes, "My dog must be homozygous (WW)." Do you agree with this conclusion? Explain your reasoning. (2 marks)

No, the conclusion is not certain.

If her dog is heterozygous (Ww), crossing with ww gives a 50% chance of a wire-haired puppy. A single wire-haired puppy could occur by chance even if her dog is heterozygous. More offspring would be needed to be confident.

- c) How would the birth of a single smooth-haired puppy have changed the certainty of her conclusion about her original dog's genotype? Explain. (1 mark)

If a single smooth-haired puppy (ww) were born, it would prove her original dog is heterozygous (Ww), because a homozygous dominant (WW) could not produce a smooth-haired offspring with a ww mate.